igate HK^+. We found that each model exhibits distinct knowledge and hallucination patterns, underscoring the need for model-specific datasets. Additionally, we showed that generic datasets are less effective at detecting model-specific hallucinations, and that model-specific datasets enable preemptive detection.

8 Limitations

Our work has a few limitations. While we evaluated three popular models, the patterns may differ in other ones. Additionally, we used only four settings to induce hallucinations given a model's correct knowledge; there may be many other ways to achieve similar aims. Finally, we only examined the two extremes of the knowledge spectrum and used one way of quantifying knowledge. Future work can investigate this middle region.

9 Ethics Statement

Our work provides insights into hallucinations that occur despite the model having the necessary knowledge. While this understanding could be misused to make models less reliable, our goal is to advance the study of the hallucination phenomenon to ultimately improve model reliability.

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A Hyper Parameters Search for Knowledge Categorization

Knowledge detection typically relies on the model's output, either through logits or generation. We focus on the generation approach, assessing whether the model consistently produces a factually correct answer among multiple samples, similar to recent work (Gekhman et al., 2024). This method is influenced by various hyperparameters including (1) number of generations, (2) sampling temperature, (3) length of generation, and (4) prompt structure.

As directly accessing factually correct is challenging, we instead examined the consistency of knowledge classification across different hyperparameter settings. A high similarity in categorization across settings would suggest comparable proximity to ground truth, reducing the impact of specific hyperparameter choices.

We evaluated the following hyperparameters:

- Shots: two different 3-shot examples and one zero-shot example.
- Temperature: $\{0.5, 1, 1.5\}$
- Number of generations: $\{5, 10\}$
- Length of generated text: $\{5, 10, 20\}$ tokens.

We started with a baseline configuration based on preliminary experiments: 3-shots, temperature of 0.5, 5 generations, and 5 tokens generated. We then modified one parameter at a time to assess its impact on classification similarity.

We categorized knowledge into three classes: "does not know" if the model did not generate the correct answer in any of the generations; "know" if the model always generated the answer; and "else" for anything in between.

We tested this approach on 1000 random TriviaQA examples across our three models. The average similarity among all 8 configurations (28 unique combinations) was 93.6% for Llama, 92.7% for Mistral, and 92.2% for Gemma, indicating a high consistency in knowledge classifications. The lowest similarity (about 80%) occurred with zero-shot configurations. Thus, indicating that the variation of configurations should not have significant effect on the knowledge classification.

Based on these results, we adopted the baseline setting as our knowledge detector, using the 3-shot prompt corresponding to the Snowballing used in subsequent hallucination classification. The high similarity between different few-shot prompts suggests that varying the few-shot examples should yield comparable results. To enhance reliability, we supplemented this approach with one greedy generation, ensuring we capture the most likely output even if temperature-based generations fail to produce it.

Note that most examples are labeled as either "know" or "does not know" and only about 5% are labeled as "else". Thus we leave the treatment of this category for future work.

One area for improvement in future research is the method of answer detection. While we use 'exact match' for simplicity and achieve relatively good results, employing methods that allow for more flexible matching could enhance recall.

B Dataset Construction Details

For TriviaQA we took 30K random examples from its training set as our initial dataset, making sure to use only examples where the answer was no longer than 5 tokens using the Mistral tokenizer. In addition, as we saw that some answers were written in uppercase, we also used the lower-case version of these answers if they contained more than 3 letters and did not contain numbers or the '/' symbol.

For the Natural Questions dataset, we also used 30K random examples, excluding examples with answers longer than 5 tokens as well as examples without an answer or with more than one answer. We again added lower-case versions of upper-case answers.

B.1 Generic Dataset Construction

To create the generic dataset, the key addition was obtaining an incorrect answer for each example. We generated these using Mistral (Jiang et al., 2023) with the following prompt:

```
Question: {question}
Correct Answer: {correct answer}
Incorrect Answer:
```

We accepted the model's greedy generation of 5 tokens as the incorrect answer if it did not contain the correct answer. During this process, we removed words such as 'Questions' or 'incorrect' that the model occasionally generated alongside the answer. For examples of hallucinated answers generated by Mistral in the generic dataset of TriviaQA, refer to Table 4.

Note that this process could potentially make the generic dataset more similar to the model-specific Mistral dataset. However, the main paper results show that this way of creating the wrong answers with Mistral provided no advantage for the generic dataset. In fact, this outcome further highlights the benefits of using a model-specific dataset. Despite the generic dataset being generated using responses from Mistral, it still performs worse than the model Mistral-specific dataset.

B.2 Specific Dataset Construction

To construct a model-specific dataset, we began with all examples from the generic dataset. We then filtered these examples based on the model's knowledge, retaining only those classified as 'known' by the model (for detailed information on this classification process, refer to Appendix A).

Subsequently, we employed a prompt-based approach utilizing Snowballing to generate responses from the model. For each example, the model produced a greedy